CREDIT CARD APPROVAL PREDICTION USING MACHINE LEARNING

PROJECT DOCUMENTATION

Submitted By

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1. ABSTRACT

The Credit Card Approval Prediction project uses Machine Learning algorithms to predict whether a customer's credit card application will be approved or rejected. The system analyzes applicant information such as income and age to make predictions. This project helps automate the approval process, reduce manual effort, and improve efficiency in banking systems.

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2. PROBLEM STATEMENT

Financial institutions receive thousands of credit card applications. Manual verification requires significant time and effort and may result in errors. This project aims to automate the approval process using Machine Learning techniques to provide quick and accurate decisions.

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3. OBJECTIVES

- To automate the credit card approval process.
- To reduce human effort and processing time.
- To improve prediction accuracy using Machine Learning.
- To demonstrate practical implementation of data science concepts.

4. FEATURES

- Automatic prediction system.
- Fast processing and response.
- Machine Learning-based decision making.
- User-friendly implementation.
- Easy to maintain and upgrade.

5. TECHNOLOGIES USED

- Python
- Pandas
- Scikit-learn
- Machine Learning
- GitHub

6. ALGORITHM USED

The project uses the Random Forest Classifier algorithm. This algorithm builds multiple decision trees and combines their outputs to improve prediction accuracy and reliability.

7. DATASET

The dataset contains applicant information such as:

- Income
- Age
- Credit Card Approval Status

The dataset is used for training and testing the Machine Learning model.

8. INSTALLATION STEPS

1. Install Python.
2. Install the required libraries using:
   pip install -r requirements.txt
3. Download the project files.

4. Run the application using:
   python app.py

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9. USAGE INSTRUCTIONS

1. Execute the application.
2. Enter the applicant's income.
3. Enter the applicant's age.
4. The system predicts whether the credit card application is approved or rejected.

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10. ADVANTAGES

- Reduces manual processing.
- Saves time.
- Improves efficiency.
- Easy to use.
- Supports future enhancements.

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11. FUTURE SCOPE

- Develop a web-based application.
- Improve prediction accuracy using larger datasets.
- Add additional applicant features.
- Deploy the system on cloud platforms.

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12. CONCLUSION

The Credit Card Approval Prediction project demonstrates how Machine Learning can automate financial decision-making processes. The system provides quick and efficient predictions, reducing human effort and improving operational efficiency.